

Musharaf Hussain Abid

AI Engineer | Agentic Systems, LLM Platforms & RAG

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years of hands-on experience across industry roles, freelancing and internships, focused on agentic systems and production LLM platforms. Builds multi-agent pipelines (LangGraph, n8n, Model Context Protocol), retrieval-augmented generation (RAG) with hybrid search and reranking, and event-driven LLM services on Azure OpenAI and serverless GPU infrastructure. Currently engineering a live enterprise contract-intelligence platform where every LLM output passes a deterministic verification layer, and cut pipeline latency by 61%. Known for validating system behavior against real production data, not assumptions.

TECHNICAL SKILLS

Agentic Systems: Multi-agent orchestration, LangGraph, LangChain, LlamaIndex, Model Context Protocol (MCP), n8n, tool/function calling, human-in-the-loop design, intent routing, workflow automation

LLM Platforms & GenAI: Azure OpenAI, OpenAI API, Groq, Llama 3, Qwen, prompt engineering, structured output parsing, LLM output verification and guardrails, fine-tuning, embeddings, evaluation

RAG & Retrieval: Hybrid search (vector + BM25), neural reranking, Azure AI Search, FAISS, ChromaDB, Qdrant, Pinecone, chunking strategy, document intelligence

Cloud & Infrastructure: Azure (Service Bus, Blob Storage, Container Apps), AWS (EC2, S3, SageMaker), Google Vertex AI, Modal, NVIDIA Triton, RunPod, Cloudflare Workers, Vercel

Backend & MLOps: Python, FastAPI, REST APIs, event-driven architecture, Docker, CI/CD, Git/GitHub, MLflow, PostgreSQL, MySQL, MongoDB

ML, Deep Learning & CV: PyTorch, TensorFlow, scikit-learn, XGBoost, Transformers, BERT, YOLO (Ultralytics), OpenCV, NLP

Languages: Python, C/C++, SQL, HTML/CSS

PROFESSIONAL EXPERIENCE

AI Engineer | Nexora AI | Remote | Jun 2026 – Present

- Own the architecture, development and production operation of LLM-based document-intelligence systems for an enterprise client, from design through live deployment.
- Design event-driven LLM services on Azure (Azure OpenAI, Service Bus, AI Search, Blob Storage) and build the retrieval, extraction and classification components behind them.
- Profile and optimize production pipelines for latency and throughput, making tuning decisions from live performance data and production logs.
- Define verification standards for LLM output so financial and date values are checked deterministically, and lead root-cause analysis of production defects.

AI Engineer | Lads Technologies | Sep 2025 – Present

- Develop and deploy AI-driven solutions in Python for production systems, including LLM-powered features and backend APIs.
- Build and integrate FastAPI services supporting scalable application delivery; contribute to code review and cross-functional task coordination.

AI Intern | ITSOLERA | Apr 2024 – Oct 2024

- Built machine learning pipelines with XGBoost for structured data and applied BERT-based models to NLP tasks on unstructured text.

Machine Learning Intern | Air University, Islamabad | Jul 2023 – Jan 2024

- Implemented a facial-recognition attendance system (MTCNN, FaceNet, TensorFlow) with automated timestamped logging, replacing manual attendance.

SELECTED PROJECTS

Contract-Intelligence Platform (Production, Enterprise Client) | Python, Azure OpenAI, Azure Service Bus, Azure AI Search

- Production LLM pipeline that extracts structured data from construction contracts and drives automated compliance alerts for a live enterprise client.
- LLM email classifier routing inbound correspondence into 8 contract-event categories using subject, body and attachments.
- Hybrid semantic-search correlation (vector search, BM25, neural reranking) linking related contractual events across a project's full email and document history.
- Cut summary-generation latency by 61% (116s to 45s) via parallel execution; deterministic verification layer on all LLM-generated dates, monetary values and durations.

WhatsApp CRM Assistant: Conversational Orchestrator | n8n, OpenAI, WhatsApp Cloud API, PostgreSQL

- Built an LLM orchestrator that extracts intent from natural-language WhatsApp messages and routes to project, lead and reminder sub-workflows, with a parallel step-by-step menu mode.
- Added stateful multi-turn flows (session state machine) and a scheduled reminder-delivery agent; migrated the data layer to an external ERP API as the source of truth.

Supplier Deals Pipeline: AI Price-Comparison Agent | n8n, Apify, OpenAI, Next.js, Vercel

- Automated product search across 6 international marketplaces, with LLM translation of queries into Chinese and Turkish and an AI-written comparison report.
- Shipped a Next.js dashboard with strict relevance filtering and automatic USD price normalization.

ImgX: Serverless GPU Image-Enhancement API | Modal, NVIDIA Triton, GFPGAN, Real-ESRGAN

- Deployed a GPU inference API on A10G GPUs; stress-tested and resolved four production bugs, including a pipe-buffer deadlock, a Triton config type mismatch and a CUDA out-of-memory failure on high-resolution inputs.

PrivateRAG: Secure Enterprise Document Assistant | LangChain, Llama 3, FAISS, Chroma

- Built a fully local RAG system so employees can query confidential documents without sending data to external APIs.

Real-Time Crowd Counting System | YOLOv8n, OpenCV, Streamlit

- Fine-tuned YOLOv8n on CrowdHuman (15,000 images), reaching mAP50 0.776 and precision 0.828, with live counting and crowd-limit alerts.

RoboMedix: Medicine Assistant for Disease Prediction (Final Year Project) | XGBoost, Bio-ClinicalBERT, Groq

- Fused XGBoost and a fine-tuned Bio-ClinicalBERT model in a hybrid classifier, with an LLM chatbot for symptom collection.

EDUCATION

Bachelor of Science in Artificial Intelligence | Air University, Islamabad

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Digital Image Processing, Reinforcement Learning.

CERTIFICATIONS

- Full Stack AI DevOps + Real Projects (AWS, Azure, GCP, Kubernetes), Udemy
- Artificial Intelligence: ML, Deep Learning, Communication, NAVTTC Prime Minister's Youth Skills Development Program (Corvit Networks)
- Operating Systems and You: Becoming a Power User, Google (Coursera)
- Operating System Foundations, Infosec (Coursera)

LANGUAGES

English (Professional Working Proficiency), Urdu (Native), Punjabi (Conversational)